

1(a)	angle (subtended at the centre of a circle) when arc (length) = radius	B1
1(b)(i)	arrow, labelled V, pointing in NE direction	B1
1(b)(ii)	arrow, labelled A, pointing in NW direction	B1
1(c)(i)	$v = r\omega$	C1
	$\omega = 0.68 / (0.093 - 0.012)$ $= 8.4 \text{ rad s}^{-1}$	A1
1(c)(ii)	$a = v^2 / r$ or $a = r\omega^2$	C1
	$a = 0.68^2 / (0.093 - 0.012)$ or $(0.093 - 0.012) \times 8.4^2$ $= 5.7 \text{ m s}^{-2}$	A1
1(d)	angular speed: same for both pieces	B1
	linear speed: less for second piece than first piece	B1
	acceleration: less for second piece than first piece	B1